NOSH ROBOTICS / COMPUTER VISION INTERN

Egg Vision Assignment

Segmentation, quantitative evaluation, and persistent-ID video tracking

Execution report | Generated 2026-10-14 11:00 UTC

SUBMISSION LIMITATION: The original assignment PDF, dataset export, SAM checkpoint, and local source video were absent. Runtime egress blocked the supplied data/video hosts, so no source HTTP response or archive check is available and no files were obtained. This report does not claim real-data metrics, dataset-wide SAM results, or an annotated copy of the supplied video.

CLASSICAL RUN	SAM INFERENCE	VIDEO EXPORT	CODE CHECKS
BLOCKED	**BLOCKED**	**BLOCKED**	**PASSED**

What the available run actually shows

The deterministic, generated 240 x 180 synthetic fixture passed the classical baseline smoke test: IoU 0.9988, Dice 0.9994. This validates basic code execution only; it is explicitly not dataset performance.

Automated verification log: 21 passed, 0 failed, 0 skipped, 0 errors; overall status: passed.

Assignment data performance

Metric	Macro / image	Pooled pixels	Meaning
IoU	N/A	N/A	Not evaluated
Dice	N/A	N/A	Not evaluated
Precision	N/A	N/A	Not evaluated
Recall	N/A	N/A	Not evaluated

Blocker recorded: FileNotFoundError: dataset directory does not exist: /home/user/nosh-robotics-egg-vision/data/egg-dataset

No evaluation scores are stated because there are no verified image annotations to score.

Methods and safeguards

1. Classical segmentation and model selection

A deterministic OpenCV baseline compares grayscale Otsu thresholding (bright/dark polarity) with Lab distance from the median image-border color. Gaussian blur and elliptical opening/closing suppress noise and fill small gaps. Nine predeclared parameter settings are ranked by macro development Dice, then development IoU; candidate order breaks any remaining tie. The predeclared grid records each candidate's design rationale, and the candidate log records development scores plus keep/discard and selection rationales. The held-out evaluation partition is read only after that selection.

2. Split and leakage safeguards

Annotated images are divided into development/evaluation at a requested 80/20 ratio (4:1). Exact decoded-image duplicates and optional regex-captured source sequences are indivisible groups; a seeded subset-sum split seeks the closest possible image ratio without group overlap. Actual counts and the seed are logged. Near-duplicate frames require suitable source grouping; no claim of automatic near-duplicate detection is made.

3. Metrics and qualitative error review

For each held-out image the runner records IoU, Dice, precision, recall, TP/FP/FN pixel counts, and an error category. Both macro per-image means and pooled pixel metrics are emitted, together with per-image CSV/JSON and four-panel input/ground-truth/prediction/error images. False negatives are blue, false positives red, and true positives green. Lowest-IoU cases are surfaced for review. A separate held-out comparison to SAM masks, when run, is explicitly reported as pseudo-label agreement and never as ground-truth accuracy.

4. Pretrained SAM outputs

The optional adapter uses official Segment Anything with a separately supplied local checkpoint. Annotation-prompt mode uses provided COCO/YOLO boxes or boxes derived from pixel masks; SAM's returned masks remain pseudo-labels. Automatic proposals are not guaranteed to be eggs. The pinned upstream README identifies the official ViT-B weight URL and Apache-2.0 model license, but its checkpoint host was not reachable and no weight or checksum was obtained. Any comparison against SAM is agreement only, not verified ground-truth accuracy.

5. Tracking

The transparent video baseline segments each frame, filters connected components, predicts centroid motion, and greedily associates boxes by distance and overlap. It writes frame labels/IDs to MP4 and a machine-readable sidecar. Crossings, occlusion, touching eggs, or poor foreground separation can switch IDs; visual review on the actual video is required.

Execution evidence and remaining work

Workstream	Status	Evidence / blocker
Classical segmentation + evaluation	Blocked	FileNotFoundError: dataset directory does not exist: /home/user/nosh-robotics-egg-vision/data/egg-dataset
Dataset/video/checkpoint retrieval	Blocked	No requests sent: runtime egress excludes the supplied Roboflow, Drive, YouTube, and SAM checkpoint hosts. No files obtained; HTTP status, redirects, MIME, and ZIP integrity are unavailable. See experiments/logs/source_access.json.
Pretrained SAM	Blocked	FileNotFoundError: dataset directory does not exist: /home/user/nosh-robotics-egg-vision/data/egg-dataset
Video tracking	Blocked	FileNotFoundError: local video not found: data/egg-video.mp4. Download the supplied video to a local file first; remote fetching is not performed.
Classical vs SAM pseudo-label agreement	Blocked	FileNotFoundError: SAM manifest not found: runs/sam/sam_manifest.json
Synthetic smoke test	Passed	Recorded controlled fixture only; not assignment-data evaluation.
Automated tests	passed	See experiments/logs/verification.json; tests include data-free fixtures and a temporary synthetic MP4.
Assignment PDF review	Blocked	No original assignment PDF was present; exact brief/grading criteria remain unchecked.

Audited environment: Python 3.11.2; 2 CPU cores; RAM available 3.6 GiB / 3.8 GiB; disk free 18.9 GiB / 20.3 GiB; GPU probe: nvidia-smi not found; no GPU confirmed. Local dataset available: False (0 image files); video: False; SAM checkpoint: False; assignment brief PDF found: False.

What is needed to finish the real-data submission

- Provide the complete assignment PDF and check for any additional requirements beyond the four tasks in the prompt.

- Place the segmentation dataset and its pixel annotations under data/egg-dataset; run the classical experiment and review the saved holdout visualizations.

- Install PyTorch/SAM, provide an official local checkpoint, run SAM over all images, and verify the manifest has no skipped images.

- Place a local copy of the specified video under data/egg-video.mp4, run tracking, inspect ID continuity, and play the exported MP4.

- Rebuild this PDF from the real logs and include only actual results; rerun the test suite before submission.

Reproducibility and disclosure

Source and output map

Path	Purpose
`configs/experiment_grid.json`	Versioned, predeclared nine-candidate tuning grid with rationales
`docs/completion_checklist.md`	Real-input blockers and exact completion steps
`eggvision/classical.py`	OpenCV segmentation candidates
`eggvision/data.py`	Dataset discovery, annotation decoding, duplicate grouping, 80/20 split
`eggvision/experiment.py`	Development-only parameter selection and held-out evaluation
`eggvision/metrics.py`	IoU, Dice, precision, recall, and error visualizations
`eggvision/pseudo_agreement.py`	Frozen holdout comparison against explicitly unverified SAM pseudo-labels
`scripts/compare_sam_classical.py`	Separate pseudo-label agreement CLI; not accuracy scoring
`eggvision/sam_inference.py`	Optional pretrained SAM image inference
`eggvision/tracking.py`	Classical detections, persistent IDs, MP4 and JSON output
`tests/`	Unit, data-format, split, evaluation, and synthetic-video tests
`experiments/logs/`	Input audit, test/smoke verification logs
`runs/`	Ignored local experiment outputs, SAM masks, and video artifacts

AI assistance disclosure

AI assistance was used to scaffold and edit the implementation, tests, documentation, experiment protocol, and report. A human must review and validate the code against the original assignment brief and actual data. The controlled synthetic fixture is clearly identified and is not represented as dataset performance. No real-data metric, SAM dataset output, or supplied-video output is claimed unless supported by its corresponding run artifact. No credentials or pretrained model checkpoint are included.

Reproduce the verified checks

```
python -m pip install -r requirements.txt
python scripts/verify.py
python scripts/run_classical.py --data data/egg-dataset --output runs/classical --seed 42
python scripts/run_sam.py --data data/egg-dataset --checkpoint models/sam_vit_b_01ec64.pth --output runs/sam
python scripts/track_video.py --video data/egg-video.mp4 --config runs/classical/selected_config.json --output runs/tracking/annotated.mp4
```